

Temporal Shuffle Null Test

News-Signal-Africa Pipeline — Methodology Note

The Question

If we completely destroy the relationship between news features and food crisis outcomes — both over time and across space — how well would the combined (AR + News) model still perform? If the answer is “almost as well as the real model,” then news is not contributing meaningful signal beyond what the autoregressive structure already captures.

What the Null Test Does

For each of 100 permutations, within each of 7 rolling CV folds:

1. Fold-aware z-scores are recomputed using only the training window — identical to the real model, ensuring no data leakage.
2. All 18 news feature columns (relative coverage + z-scores) are **double-shuffled**. The 4 AR features and the target are left completely untouched:
 - *Temporal shuffle*: for each district independently, its news observations are randomly reordered across time periods. This breaks the link between news timing and crisis timing within each district.
 - *Spatial shuffle*: for each time period independently, news values are randomly swapped across districts. This breaks the link between which district had which news and which subsequently had a crisis.
3. The **combined model only** (AR + shuffled news) is trained on the shuffled data with fixed hyperparameters (300 iterations, no early stopping) and its PR-AUC is recorded on the test fold. The AR-only model is not retrained in the null loop — it appears only as a reference from the real results.
4. PR-AUC is averaged across all 7 folds to produce one null draw. This is repeated 100 times, yielding a null distribution of 100 draws.

Results

Null distribution (combined model, shuffled news): 0.795 ± 0.009 (range: 0.772 – 0.818, $n = 100$ permutations)

Real AR-only PR-AUC (observed, not shuffled): 0.800

Real combined model PR-AUC (observed, not shuffled): 0.823

p-value: 0.00 (0 of 100 null draws $\geq$ real combined model)

Interpretation

Even after completely scrambling which news happened where and when, the combined model still scores 0.795 PR-AUC. This is because the AR features (IPC lag, 2-year persistence, spatial lag) carry most of the predictive power — and they are never shuffled. The real combined model scores 0.823, an increment of +0.028 above the null. Since zero of 100 null permutations reached this

level, the news contribution is statistically significant ($p \leq 0.01$) and is not an artefact of distributional patterns or model capacity.

Note: the null distribution covers the combined model only. The AR-only value (0.800) is shown as an additional reference to illustrate that real news adds +0.023 over AR-only, and that even shuffled news (0.795) falls only marginally below AR-only — confirming that the combined model's gain is driven by temporal and spatial news signal, not by the mere presence of additional features.